



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review Updated Article for IEEE Access Access-2024-31262

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Wed, Aug 21, 2024 at 5:09 AM

Reply-To: chiyuan.chen@ieee.org

To: khare.pranav@gmail.com

21-Aug-2024

Dear Mr. Khare:

Recently you reviewed an article submitted to IEEE Access entitled, "Optimized Data-Flow Integrity for Modern Compilers". Please note that this article was rejected, and the authors were asked to update their article based on the reviewers' concerns.

The authors have resubmitted their revised article and supplied a "list of updates" that highlights all edits made, as well as their responses to the previous reviewers' concerns.

We would appreciate if you could take another look at this improved version to determine if your concerns have been addressed, and to ensure that you are satisfied with the updates.

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Once you accept my invitation to review this manuscript, you will be notified via email with directions on how to access the manuscript and the reviewer instructions in the Reviewer Center.

Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Chi-Yuan Chen
IEEE Access Associate Editor
chiyuan.chen@ieee.org, chency@niu.edu.tw

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Optimized Data-Flow Integrity for Modern Compilers

AUTHORS: IRENE, DÍEZ-FRANCO; Ugarte-Pedrero, Xabier; Bringas, Pablo G.

ABSTRACT: Non-control-data attacks are those attacks that purely target and modify the non-control data of a program, such as boolean values, user input or configuration parameters, and leave the control flow of a program untouched. These attacks were considered a niche due to the high difficulty in crafting attacks that do not modify the control flow.

However, in recent years researchers have already demonstrated that non-control-data attacks can be automatically constructed and that they pose a significant threat because they can compromise critical and widely used software, such as web browsers and the Linux kernel. Moreover, they can also be used to disable or bypass state-of-the-art software security techniques, such as control-flow integrity. The most promising technique to protect against non-control-data attacks is data-flow integrity, however, modern compilers do not implement this protection yet. In this work we present an optimized data-flow integrity implementation for modern compilers that reduces the amount of basic blocks that need to be protected in an average of 45.8%, it also has broader security guarantees due to its more precise static analysis. Finally, we evaluate the completeness of our optimized data-flow integrity implementation.

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